Imtiaz Ahmed

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EDUCATION

The Cooper Union for the Advancement of Science and Art 2029

B.E. – Mechanical Engineering
GPA: 3.975/4.000
Relevant course work: Eng. Design and Problem Solving, Principles of Design, Intro to Computer Science
Scholarships: Half-Tuition Scholarship, Innovator Scholarship

RELEVANT EXPERIENCE

Ecolibrium at Loisaida, Manhattan, NY January 2026 – Present
Sensor Case Engineering Intern

- Designed 3D-printed wireless air quality sensor enclosure to improve ESP32 display and sensor integration
- Developed electronics tray/outer shell architecture to facilitate fast and jig-less assembly and easy maintenance
- Selected power electronics to minimize electrical intermittencies and brown-outs upon unplugging chargers
- Manufactured silicone sensor sealing gaskets to minimize sensor exposure to internal airflow
- Calibrated sensors with respect to evaluation sensor units and mercury thermometers to determine optimal self-heating offsets
- Reduced mechanical assembly times from over 10 minutes to under 3 minutes
- Designed and manufactured 10 laser-cuttable enclosure prototypes for educational purposes
- Iterated laser-cut design for manufacturability and designed hinged jig to facilitate assembly
- Monitored component procurement and costs in Excel

Cooper Union Motorsports, Manhattan, NY September 2025 - Present
Chassis Subsystem Member

- Designed fully parametric space-frame FSAE chassis in Solidworks to facilitate rapid modifications to accommodate suspension and engine hardpoints
- Incorporated Formula SAE 2026 dimensional, material, and packaging constraints into chassis design
- Arranged driver and engine placement to optimize center of gravity
- Performed FEA in ANSYS Mechanical, applying suspension loads and evaluating chassis deformation to determine 1500 Nm/deg torsional rigidity
- Built a full-scale PVC/duct-tape chassis prototype to evaluate driver egress, component packaging, and dimensional constraints before fabrication
- Prepared chassis for manufacturing through determining tube miter geometry, creating pipe cut list and Excel BoM, and relevant engineering drawings

Cooper Union Satellite Initiative, Manhattan, NY September 2025 - Present
ADCS Subteam Co-lead

- Designed a mechanical magnetorquer test rig integrating the relevant test components into a compact cube-shaped assembly
- Developed a water-supported testing concept to minimize friction during one-dimensional magnetorquer motion and isolate the actuator's response
- Investigated multiple concepts for a three-axis magnetorquer test rig, and performed initial calculations for an aerostatic bearing-based test rig to determine necessary supply pressures and nozzle geometries
- Analyzed the existing Simulink ACDS model, tracing signal paths between environmental models, sensors, control logic, and actuators to produce documentation for new members

Cooper Union Bio-inspired Robotics Design Lab, Manhattan, NY February 2026 - Present
Research Assistant

- Investigated membrane attachment and tensioning methods for PLA flapping-wing structures
- Designed and prototyped a hand-operated friction-based membrane-tensioning mechanism to evaluate an alternative method for controlled membrane placement
- Evaluated the prototype through rapid testing and determined that the mechanism did not provide sufficient performance for the intended application
- Documented membrane manufacturing and attachment procedures to support repeatable wing fabrication

Cooper Air Design-Build-Vertical Flight Team, Manhattan, NY January 2026 – March 2026
Mechanical Subteam Member

- Designed battery mount to cleanly interface with power distribution board and existing attachment points
- Designed a LiDAR mounting system that minimized modifications to existing UAV frame plates
- Incorporated tool access constraints to allow efficient installation and removal during competition

PROJECTS

Linkage-actuated Robotic Arm

- Designed and prototyped a robotic arm using coupled four-bar and bell-crank mechanisms with Arduino-controlled 9g servo actuation
- Developed a mechanically coupled elbow and claw system in which passive linkage geometry transfers motion between joints
- Designed a bell-crank and scissor-link claw mechanism to convert servo rotation into gripping motion while maintaining a compact actuator arrangement
- Used OnShape assembly kinematics to characterize the relationship between elbow and claw angles and derive software compensation functions for maintaining open and closed claw states across elbow positions
- Implemented potentiometer-based manual control through Arduino and constrained servo commands to mechanically valid 0–180° operating ranges
- Iterated mechanical geometry and actuator placement to reduce moving mass and improve torque transmission; identified servo torque limitations as primary constraints on motion smoothness
- Moved center of gravity 40% closer to the base at maximum extension compared to a similar arm without linkages, a difference that would increase even more with heavier servos

Ping Pong Ball Circulation System

- Designed and prototyped a closed-loop ping-pong ball handling system integrating a flywheel launcher, separator wheel, infrared sensing array, and automated ball-recovery mechanism using Arduino-based control
- Developed a single-flywheel launcher with an adjustable launch angle and modular attachment points; iterated through three launcher prototypes to improve launch consistency, feed reliability, and maintainability
- Designed a separator-wheel feed mechanism to regulate ball input and reduce flywheel momentum loss during consecutive launches; added a 3D-printed anti-jam cover and vibration motor to improve feed reliability
- Developed a velocity-measurement system using 16 infrared break-beam sensors arranged across two sensing planes; used beam timing to measure ball velocity and track circulation rate
- Designed the recovery subsystem using a curved backboard, netting, gravity-assisted routing, and a DC motor-driven propeller to return balls through PVC tubing to the launcher
- Iteratively tested and modified mechanical geometry to reduce ball bounce-outs, feed jams, and launch-height variation; achieved 63 circulations in 60 seconds during final demonstration

VEX Ball Launching Robot

- Designed and built a side-feeding ball-launching VEX robot integrating a dual-roller intake, conveyor transport, and mechanically actuated ball-launching system controlled through RobotC teleoperation
- Designed a vertical ball-transfer system using a laser-cut wood channel and custom 3D-printed mounting brackets, routing balls from the intake to the launcher while maintaining alignment
- Developed an elastic-powered rack-and-pinion launching mechanism to convert stored elastic energy into linear ball acceleration and launch balls through a side-mounted exit
- Created a complete Onshape assembly to integrate VEX structural components, custom-fabricated parts, drivetrain, intake, conveyor, and launching mechanisms

Color-based Marble Sorter

- Designed and prototyped an automated marble classification and routing system using VEX hardware, reflected-light sensing, and servo-actuated mechanisms
- Developed a mechanical separation and feeding system using a 3D-printed rotating gate to isolate individual marbles and reduce interference during sensor measurements
- Implemented threshold-based classification logic in RobotC, calibrating reflected-light sensor ranges through repeated testing to distinguish five marble categories
- Designed a servo-driven routing mechanism using a five-position rotating collection assembly to direct classified marbles into category-specific containers
- Integrated sensor acquisition, marble release timing, and servo positioning into a sequential automated sorting cycle; achieved reliable classification of 4/5 marble categories
- Diagnosed failure modes including double-feeding, ambient-light sensitivity, and overlapping sensor response ranges; identified controlled illumination and improved calibration as potential paths to improve classification reliability

Ecolibrium 3D-printed Indoor Air Quality Monitor

- Developed enclosure architecture to integrate ESP32 display, Sensirion SEN66 environmental sensor, and power electronics while maintaining sensor exposure to ambient conditions
- Iterated between laser-cut acrylic and FDM-printed PLA architectures based on manufacturing, assembly, thermal, and deployment requirements
- Designed a modular electronics tray and separate outer shell to simplify assembly, servicing, and future hardware changes
- Designed all electronic components to interface with skeleton with dovetails, snap-fits, or slide-fits to eliminate all adhesives and screws
- Used J5019 charging/voltage boost board to eliminate brown-outs upon unplugging charger and remove internal USB-A and micro USB cables for charging ports

SKILLS

CAD & Design: SolidWorks, Onshape, Fusion 360, AutoCAD, Engineering Drawings
Programming & Simulation: Python, MATLAB, ANSYS, Arduino/C++, Simulink
Fabrication & Manufacturing: 3D Printing (FDM), Laser Cutting, Manual Lathe, Manual Mill
Electronics: Soldering, Electrical Measurement, Sensor Integration
Tools: Excel, Git/GitHub, Notion